

Multicalibration for Unbiased Prevalence Estimation

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Abstract

Political scientists increasingly use classifiers, and now large language models (LLMs), to estimate how prevalent concepts are in natural language, such as topics in legislative agendas or categories in open-ended survey responses. Standard validation (reporting accuracy or AUC on a labeled sample) certifies neither valid prevalence estimates nor their cross-context comparability. When the target differs from the validation sample (covariate shift), an accurate classifier can still produce badly biased prevalence estimates, and the usual remedies redo work for every new target. We show that *multicalibration*, calibration conditional on input features rather than on average, is sufficient for unbiased prevalence estimation under any covariate shift along the calibrated features, yielding an instrument calibrated once and applied to any target within its support. A simulation and two applications, cross-national LLM topic coding and a controlled employment benchmark, show near-zero bias under shift, where standard methods miss by a large margin. The correction also works on binary model outputs without the need for eliciting probabilities.

Introduction

A growing share of empirical political science depends on machines reading text. Researchers use automated classifiers, and increasingly large language models (LLMs) for this task. Some recent examples of such applications are coding policy agendas across countries (Baumgartner et al. 2006; Benoit et al. 2026), detecting protest and conflict events in news (Overos et al. 2024; King and Lowe 2003), scoring democratic institutions (Weidmann et al. 2026), categorizing open-ended survey responses (Mellon et al. 2024), and labeling social media at a scale that hand-coding cannot reach (Gilardi et al. 2023; Ornstein et al. 2025). In many of these studies the quantity of scientific interest is not the label on any single document but an *aggregate*: how prevalent a category is in a population. What fraction of a parliament’s questions concern law and crime? How often does a topic appear in one country’s media versus another’s? These are *prevalence* (or *quantification*) problems (González et al. 2017; Hopkins and King 2010).

A common playbook for such measurement designs is to validate the model on a labeled sample of data and then apply it to other datasets. Commonly, researchers validate an automated coder by its discriminative performance (accuracy, F_1 , AUC, or agreement with human coders) on the labeled validation sample (Grimmer et al. 2022; Halterman and Keith 2025). A coder that clears a high bar is then applied to new corpora: other countries, earlier decades, different document types (Benoit et al. 2026; King and Lowe 2003). This practice rests on an implicit assumption: that performance

established on the validation sample transfers to the populations where the instrument is actually used. That assumption is rarely examined, and for prevalence estimation it is frequently wrong.

The problem is that discrimination and prevalence accuracy are different properties. A classifier can rank documents almost perfectly (separating positives from negatives) while systematically over- or under-stating *how many* positives there are. Worse, the direction and size of that prevalence error depend on the composition of the target population. When the target differs from the validation sample in its feature distribution (a *covariate shift*: the mix of languages, document types, eras, or authors changes, but the meaning of the category does not), a coder that produced unbiased prevalence estimates on the validation sample can produce badly biased ones elsewhere. This covariate shift arises routinely whenever a single instrument is applied across corpora that differ in language, venue, era, or author. The shift we should expect here is covariate shift: what changes across corpora is which documents appear, while the rule mapping a document’s content to its category stays fixed. That is the opposite of the label-shift assumption behind classical quantifiers, which requires documents of a given category to look alike across corpora, implausible across languages and venues. When it is present, covariate shift is a first-order threat to measurement validity (Adcock and Collier 2001) that discriminative validation does not detect.

Political methodology already offers tools that engage parts of this problem, but none delivers a *reusable* instrument. Quantification methods such as ReadMe (Hopkins and King 2010) and its successor (Jerzak et al. 2023) estimate category proportions directly, but under a *label-shift* assumption that is the wrong invariance for the covariate-shift regime we study. Design-based supervised learning (Egami et al. 2023) and prediction-powered inference (Angelopoulos et al. 2023) debias a downstream estimate by combining cheap machine labels with a small set of gold labels *drawn from the target sample itself*, powerful and assumption-light, but target-specific by construction: each new population requires its own gold sample. Importance weighting (Shimodaira 2000; Sugiyama et al. 2007) is the classical correction for covariate shift, but it too must be redone for each target.

We show that *multicalibration* fills this gap. A device is multicalibrated when it is calibrated conditional on input features, not merely on average (Hébert-Johnson et al. 2018). A multicalibrated device, calibrated once on a labeled source sample, returns unbiased prevalence estimates on any target whose feature distribution has shifted, provided the shift is captured by the calibration features and the target lies within the calibration support (Kim et al. 2022). No target-specific re-estimation is required. We develop this into a measurement protocol for prevalence estimation from political text, and in doing so clarify which correction suits which setting: quantification and calibration methods are each valid under a particular regime of distribution shift.

Measurement, quantification, and covariate shift in text-as-data research

Extracting quantities from political text has a long lineage, from scaling policy positions with words as data (Laver et al. 2003) to automated event extraction matching human coders (King and Lowe 2003) to the supervised and unsupervised methods synthesized by Grimmer et al. (2022) and Grimmer and Stewart (2013). Throughout, the validation standard has been agreement with human coders: intercoder reliability for the human benchmark itself (Hayes and Krippendorff 2007), and accuracy or AUC for the machine. LLMs are the newest entrant, and a rapidly growing literature evaluates them as zero-shot coders of political concepts (Halterman and Keith 2025; Benoit et al.

2026; Ornstein et al. 2025; Weidmann et al. 2026), generally on the same discrimination metrics. Our argument is that for the prevalence estimands these studies target, agreement-on-average in a validation corpus is necessary but not sufficient: it does not certify that estimates transport to the populations of interest.

The task of estimating class prevalence from imperfect classifiers is *quantification* in machine learning (González et al. 2017) and is the explicit target of ReadMe (Hopkins and King 2010) and ReadMe2 (Jerzak et al. 2023) in political science. These methods estimate category proportions without committing to per-document labels, and ReadMe2 improves robustness to changes in the document-feature distribution. The applied default is Classify & Count, reporting the fraction of documents the coder labels positive. Its bias-corrected relatives (Adjusted Count, i.e. the Rogan-Gladen adjustment (Rogan and Gladen 1978); the Saerens-Latinne-Decaestecker EM algorithm (Saerens et al. 2002); distribution-matching methods (Maletzke et al. 2018)) and the ReadMe family, however, all rely on *label shift*: that documents of a given category look alike across corpora, so only the category’s overall frequency changes from one corpus to the next. In much comparative text-as-data that invariance is implausible, because the same category is expressed with different words across languages, venues, and eras. The regime we study is the opposite, *covariate shift*, where the mix of documents changes but the coding rule does not; we define both formally in the next section. Real shifts are often mixtures, but to the extent a covariate-shift component is present, label-shift quantifiers are not guaranteed to be unbiased, as our results confirm.

A complementary literature corrects downstream estimates computed from imperfect machine labels. Design-based supervised learning (Egami et al. 2023) and prediction-powered inference (Angelopoulos et al. 2023) combine many cheap predictions with a small probability sample of gold labels to produce design-consistent estimates with valid uncertainty. These methods are assumption-light (they need no covariate-shift or ignorability assumption) but they are *target-specific*: the gold sample must come from the target population, so each new country, period, or subgroup requires fresh hand-coding. We treat these methods as complementary and return to the trade-off in the discussion. Related work on learned proxies in causal inference (Knox et al. 2022) and on the reliability of LLM annotations (Bisbee et al. 2024; Argyle et al. 2023; Rodriguez and Spirling 2022) shares our concern that imperfect measures can distort inference, but does not address the transportability of prevalence estimates across populations. Closest in spirit, Weidmann et al. (2026) find that LLM coders of democracy indicators deviate from expert coders in systematic, model-specific directions (some consistently over-, others under-stating), which is close to the feature- and context-dependent error that a feature-conditional correction is designed to remove.

Calibration (that predicted probabilities match empirical frequencies) is the property that links a probabilistic coder to unbiased prevalence: if a device is globally calibrated on a population, the mean of its scores equals that population’s prevalence. But global calibration is a property of one population and need not survive a shift. Empirically, LLM calibration degrades two- to threefold under language shift even when accuracy is preserved (Yang et al. 2023) and varies widely across tasks (Ren et al. 2025). Confidence-elicitation methods (verbalized confidence (Tian et al. 2023), token log-probabilities (Kadavath et al. 2022), self-consistency (Wang et al. 2023)) and post-hoc recalibration (e.g., isotonic regression (Wu and Resnick 2024)) target *global* calibration and so inherit this non-transfer. Multicalibration (Hébert-Johnson et al. 2018; Detommaso et al. 2024), by contrast, conditions on features and is therefore stable under reweighting of those features; this is the property we exploit.

Prevalence Bias Reduction through Multicalibration

Consider a binary outcome $Y \in \{0, 1\}$, features X , and a device $h(X) \in [0, 1]$. The estimand is the prevalence $\pi^* = P^*(Y = 1)$ in a target population, to be estimated from unlabeled target data $\{X_i\}$ and the device h . We assume *covariate shift*: $P(X)$ differs between calibration and target populations while $P(Y | X)$ is stable. Substantively, the coding rule is fixed: a document with given content receives the same label whatever corpus it came from (what a codebook is meant to guarantee), while the mix of documents changes. This is the complement of the *label-shift* assumption behind classical quantifiers, which instead holds $P(X | Y)$ fixed while the prior $P(Y)$ moves.¹ Under concept drift, where $P(Y | X)$ itself changes across contexts, no purely statistical correction recovers the target prevalence without labels drawn from the target.

A device is *globally calibrated* on a population if $\mathbb{E}[Y | h(X) = p] = p$ for all p . Under global calibration, $\mathbb{E}[h(X)] = \mathbb{E}[Y] = \pi$, so averaging predictions estimates prevalence without bias. But calibration on one population does not imply calibration on another. Partition the population into feature-defined groups G with weights w_G , and let $\epsilon_G = \mathbb{E}[h(X) - Y | X \in G]$ be the device’s mean signed error (predicted minus true) in group G . Global calibration requires only that these errors cancel *on average*: $\sum_G w_G \epsilon_G = 0$. Under covariate shift the weights become w_G^* , and the prevalence bias is

$$\hat{\pi}^* - \pi^* = \sum_G w_G^* \epsilon_G,$$

which is generally nonzero unless every $\epsilon_G = 0$. Because ϵ_G is estimated on the source, the identity is exact only when each group is homogeneous enough that its composition is the same in the source and the target (equivalently, the density ratio is constant within each group), so that all shift occurs across groups and none within them. With coarser groups whose within-group mix also changes, the target’s true within-group error departs from the source estimate ϵ_G , and the identity becomes a first-order approximation. A device can be perfectly calibrated on the validation sample (its group errors offsetting under w_G) and badly biased on a target that reweights those same groups. High AUC offers no protection: AUC depends only on the ranking of scores, so it is invariant to all miscalibration, including the feature-conditional errors ϵ_G that drive prevalence bias under shift: a classifier can have near-perfect AUC and still be badly biased in prevalence.

The bias decomposition shows why existing methods fail to produce unbiased estimates. The corrections that operate on a device’s outputs (Classify & Count, the Rogan-Gladen adjustment, and global recalibration by isotonic regression (Wu and Resnick 2024)) reduce to reweighted group errors: the aggregate positive and error rates they estimate on the source are weighted averages over groups that no longer cancel once the target reweights those groups. The label-shift quantifiers (SLD/EMQ (Saerens et al. 2002), distribution-matching methods, and the classifier-free ReadMe family (Hopkins and King 2010; Jerzak et al. 2023)) reach the same failure by a different route: having no per-document score, they carry no ϵ_G to reweight, but the identity they solve presumes $P(X | Y)$ is stable.

The fix is to bring *every* ϵ_G towards zero, not just their weighted sum. This can be achieved via *multicalibration*. A predictor f is *multicalibrated* with respect to a class of groups $\mathcal{G}$ if $\mathbb{E}[Y | f(X) = v, X \in G] = v$ for all $G \in \mathcal{G}$ and values v (Hébert-Johnson et al. 2018).² A multicalibrated predictor

¹The heuristic that stable $P(Y | X)$ is natural when features cause the outcome, and stable $P(X | Y)$ when the outcome causes the features, traces to @scholkopf2012causal; we rely on the invariance rather than on adjudicating the causal direction.

²Technically, a slightly weaker property, *multi-accuracy* (zero average error within each group, rather than at every

delivers the Kim et al. (2022) “universal adaptability” guarantee: for any target whose density ratio dP^*/dP is measurable with respect to $\mathcal{G}$ (equivalently, whose reweighting is captured by the calibration features),

$$\mathbb{E}^*[f(X)] = \mathbb{E}\left[\frac{dP^*}{dP}(X) f(X)\right] = \mathbb{E}\left[\frac{dP^*}{dP}(X) Y\right] = \mathbb{E}^*[Y] = \pi^*,$$

where the middle equality holds because the density ratio is constant within each $\mathcal{G}$ -cell, so the within-cell error $\epsilon_G = 0$ carries through. One instrument, calibrated once on source data, is thus unbiased on every target whose shift lies along the calibrated feature dimensions and within support, with no knowledge of the shift and no target-specific estimation.

The classical way to exploit the same density ratio is *importance weighting*: estimate dP^*/dP from unlabeled source and target samples and reweight the labeled source data to match the target (Shimodaira 2000; Sugiyama et al. 2007). It is valid under the same covariate-shift assumption, but it corrects the *sample* rather than the *predictor*, so the density ratio must be re-estimated for every new target, and it breaks down when the target places mass where the source has almost none (a positivity violation). Multicalibration instead folds the correction into the predictor once, so one calibrated instrument transports to every in-support target without per-target reweighting. Matching per-target importance weighting without per-target work is the practical payoff we return to in the applications.

The unbiasedness through multicalibration guarantee rests on two distinct requirements: First, *support/overlap*: the target’s feature values must lie within the calibration support, so that there is calibration data on which the predictor was corrected. The second, and more demanding, is that the calibration feature class $\mathcal{G}$ be *expressive enough that within-cell $P(Y | X)$ is stable across populations*: that the features capture every dimension along which the shift interacts with the outcome. The support condition is the analogue of the positivity that importance weighting needs; the expressiveness condition is an ignorability assumption discharged *once*, at calibration time, rather than re-estimated per target. In text settings the expressiveness condition is the more fragile one: coarse metadata cells (the same “newspaper” document type in two eras with shifted writing style) may share all observed feature values while differing in latent content that drives Y , so that within-cell $P(Y | X)$ is not in fact stable. The natural mitigation is to enrich $\mathcal{G}$ with text-derived features (document embeddings (Rodriguez and Spirling 2022)) so that the calibration cells track semantic content, not just metadata. When either condition fails (a novel feature value, or latent variation no feature captures), estimates degrade, as our out-of-support results show.

In practice we obtain a multicalibrated predictor with MCGrad (Tax et al. 2026), a post-hoc gradient-boosting multicalibrator.

Simulation

We illustrate the mechanism in a setting simple enough to verify by hand. We generate synthetic data with a binary covariate $X \in \{0, 1\}$ and outcome Y with $P(Y = 1 | X = 1) = 0.85$ and $P(Y = 1 | X = 0) = 0.15$. A classifier produces deterministic, systematically biased scores (a 10% multiplicative bias within each stratum, so $\hat{p} = 0.135$ when $X = 0$ and $\hat{p} = 0.935$ when $X = 1$), wrong within each stratum but corrected to the true prevalence by a single global recalibration

score level within each group), is already sufficient for unbiased prevalence; multicalibration is stronger but the better choice in practice. We discuss the distinction, and why, in SI Section S1.8.

step on the balanced training distribution $P(X = 0) = 0.5$. We estimate prevalence on target distributions with $P(X = 0)$ ranging from 0.01 to 0.99, holding all calibration parameters fixed at their training values, over 50 replications.

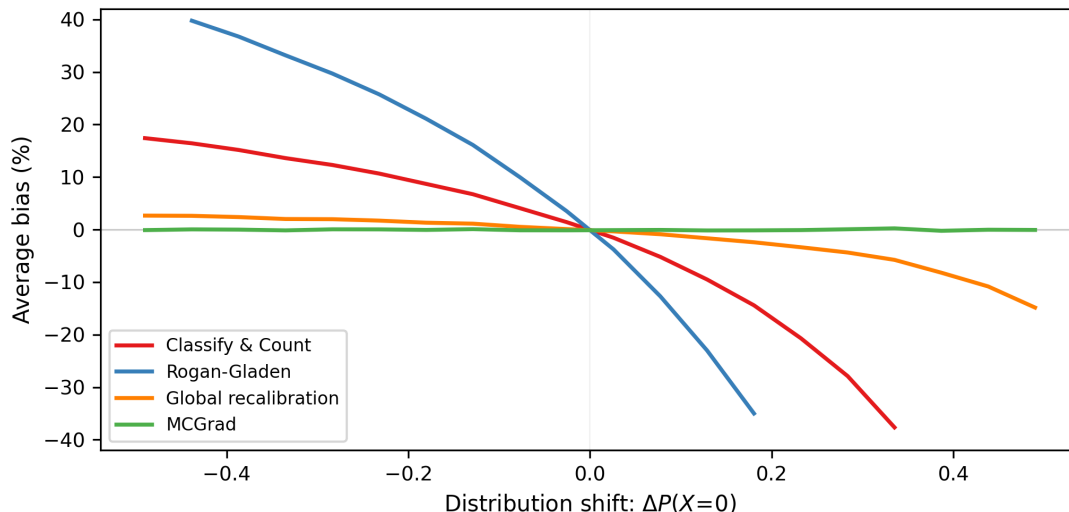


Figure 1. Relative prevalence bias under covariate shift (50 runs). The x-axis is the change in $P(X = 0)$ from the training value of 0.5. Classify & Count and Rogan-Gladen diverge with shift; global recalibration (multiplicative, the method of SI Section S1.6) shows moderate bias; the multicalibrated estimator stays near zero. Cropped at $\pm 40\%$; see SI Figure S2 for the full range and additional methods.

At the training distribution ($\Delta P(X = 0) = 0$) all methods are approximately unbiased (Figure 1). As the target shifts, they diverge: Rogan-Gladen is most unstable, its ratio form amplifying error beyond $\pm 40\%$; Classify & Count grows in the same direction; global recalibration shows moderate but nonzero bias, up to roughly 15% at the most extreme shift. The multicalibrated estimator (here, stratum-specific additive corrections) maintains near-zero bias.³

Cross-national LLM coding under covariate shift

Our main application is the modal text-as-data workflow: one coding instrument applied across countries and languages. We use Claude Opus 4.6 to classify 30,000 Comparative Agendas Project texts (Baumgartner et al. 2006) for whether they concern Law & Crime (CAP major topic 12), scored against the published CAP expert codes as ground truth. The documents span six sub-populations across four countries and languages, 5,000 each: Danish parliamentary questions, Spanish oral questions, U.S. congressional bills, Belgian newspaper articles, Spanish media articles, and Belgian TV news. The estimand is the Law & Crime share in each corpus, and the design reproduces the situation a researcher faces: code a concept once, then estimate its prevalence across many corpora that differ in language, venue, and format. This spread of sub-populations is what lets us trace bias across the shift gradient.

We compare two LLM output modes spanning current practice. The first is the discrete Yes/No classification used in essentially all applied studies. The second is a probability score obtained

³To confirm that bias reduction does not come at the cost of variance, we show RSME results in the appendix SI Figures 23–23

by direct elicitation, where the model estimates $P(\text{Yes})$ without first committing to an answer, representing the confidence-elicitation literature’s recommendation (Tian et al. 2023; Kadavath et al. 2022). Discrimination is strong in both modes: AUC 0.960 (binary) and 0.987 (scores), with per-language AUCs of 0.983–0.994. The calibration set ($n \approx 13,400$) draws equally from four sub-populations, ensuring all four countries and languages are represented. Two sub-populations are held out as out-of-support targets (Spanish media and Belgian TV), which share country and language with the calibration data but introduce an unseen document type. We calibrate MCGrad on observable metadata only: country, document type, party, decade, and document length. These features play the role of the group class $\mathcal{G}$ in the guarantee above: fitting MCGrad on them makes the predictor approximately multicalibrated with respect to every group that can be formed from them (any subset or intersection of their values), up to finite-sample error. This is what makes the prevalence estimate unbiased on any target whose reweighting is captured by these features. For the binary condition, MCGrad receives the Yes/No label as a categorical feature with scores initialized at the calibration base rate; it never sees a probability from the model. For the score condition, it receives $P(\text{Yes})$ directly. We benchmark against Classify & Count, Rogan-Gladen, isotonic regression, importance weighting (IPW), and SLD (SI Table S2).

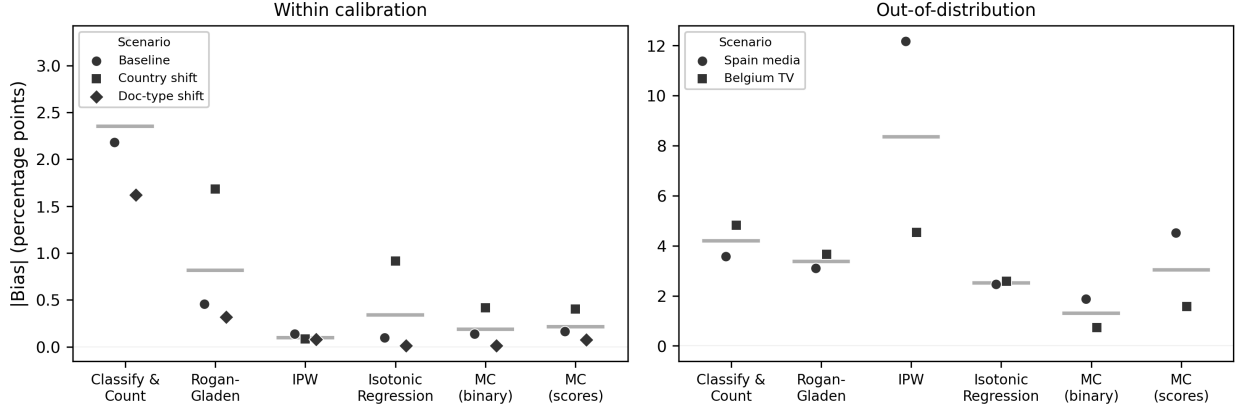


Figure 2. Absolute prevalence bias (percentage points) across the shift gradient: a no-shift baseline, two within-support shifts (country and document-type composition), and two out-of-support targets. Marker shape denotes scenario; horizontal lines are scenario means. The two multicalibration variants, MCGrad on bare Yes/No labels (MC binary) and on elicited probability scores (MC scores), are adjacent for direct comparison; within support they are indistinguishable (both $\leq 0.4\text{pp}$, a difference smaller than the roughly 0.4–0.6pp sampling error in the gold base rate), and the headline finding is that the binary variant requires no probability output at all. Full numbers in SI Table S2.

The failure pattern is clear (Figure 2). Classify & Count (counting the model’s Yes labels, the default in applied work) is biased by +1.6 to +4.8pp across all scenarios. Rogan-Gladen reduces this within support but still misses by +3.1 to +3.7pp out of support. Isotonic regression is good within support ($\leq 0.9\text{pp}$) but worse out of support (2.5–2.6pp). SLD, the label-shift quantifier we benchmark, is already biased at the no-shift baseline (+7.4pp, reflecting that the elicited scores are not the calibrated posteriors its EM step assumes) and the shift amplifies this to +21.6pp out of support, confirming that the label-shift family does not transport under covariate shift.⁴

⁴ReadMe [hopkinsking2010nonparametric], the canonical political-science quantifier for this kind of problem, is not a direct competitor to the methods in Figure 2: it is classifier-free, estimating the proportion from the raw document text rather than correcting the LLM’s outputs, and its bias runs far off their scale. Because it is nonetheless the tool a political-science reader will expect, we ran the comparison and, to keep the main text focused, report it in SI

Multicalibration on the bare Yes/No labels achieves near-zero bias within the calibration distribution ($\leq 0.4\text{pp}$) and degrades gracefully out of support: $+0.7\text{pp}$ on Belgian TV and -1.9pp on Spanish media. This is obtained from the Yes/No label and metadata alone, with no probability scores. Multicalibration on probability scores matches it within support ($\leq 0.4\text{pp}$) but is somewhat worse out of support (-4.5pp on Spanish media, $+1.6\text{pp}$ on Belgian TV). That bare labels match or beat elicited probabilities is our most practically important finding: it is the metadata features and the calibration sample, not the quality of the model’s confidences, that drive the correction. IPW achieves near-zero bias within support but fails severely out of support (-12.2pp on Spanish media) because the target contains feature values absent from the source (a positivity violation), and, unlike multicalibration, it must re-estimate density ratios for each target.

Two caveats temper the within-support numbers. First, the gold standard is itself the CAP expert coding, so “true” prevalence is an estimate: the binomial sampling error on a sub-population base rate (5,000 documents) is about $\pm 0.4\text{--}0.6\text{pp}$ ($\pm 0.8\text{--}1.1\text{pp}$ at 95%), so sub-1pp differences within support should be read as ties rather than rankings, and the substantive comparison is between near-zero-bias methods (multicalibration, IPW, isotonic within support) and the multi-point biases of Classify & Count and the label-shift quantifier. We report this binomial term rather than the full-pipeline bootstrap (resampling the calibration set, refitting, and recomputing) because here it dominates: with a 13,400-document calibration set the calibration-fitting variance is small relative to the gold-standard sampling error, and the ranking of methods by point bias is robust to it; the ACS benchmark, where truth is exact, exercises the full bootstrap, whose 200-iteration RMSE closely tracks the absolute bias in every scenario (Table S1). Second, the field setting lacks an oracle by construction; the controlled benchmark below supplies the identification-clean confirmation where truth is known exactly. We replicate the entire analysis with an open-weight model, Llama 3.3 70B, in SI Section S2: the pattern holds (MCGrad $\leq 0.2\text{pp}$ within calibration), confirming the findings are not specific to a proprietary model and are reproducible without API access.

Employment prevalence with exact ground truth

As a check with exact ground truth, we estimate employment prevalence from American Community Survey microdata. The true rate in any subpopulation is known, and the classifier is an ordinary logistic regression rather than an LLM, so this confirms the correction is not specific to language models or to noisy gold labels. We predict employment from 16 sociodemographic features, training on eight states (2016–2018; approximately 1.5M observations) and calibrating on a held-out set ($n \approx 644,000$). Because employment rates vary sharply by age (76% for ages 25–54 versus 17% for 65+), we construct covariate shifts by resampling the test set to be young-skewed, old-skewed, or bimodal, yielding true employment rates from 12.8% to 46.0%; all calibration parameters are fixed across scenarios. We evaluate both on in-distribution states and on six held-out states.

Section S7 (Table S4). It fails as the label-shift diagnosis predicts, now shown directly for the canonical method rather than inferred from SLD alone: near-zero only without shift, and badly biased and highly unstable under it, by tens of percentage points and exceeding $+60\text{pp}$ on some shifts. Its successor ReadMe2 [@jerkkingstrezhnev2023improved] refines the feature summary but estimates proportions under the same label-shift assumption, so in principle it cannot escape this failure.

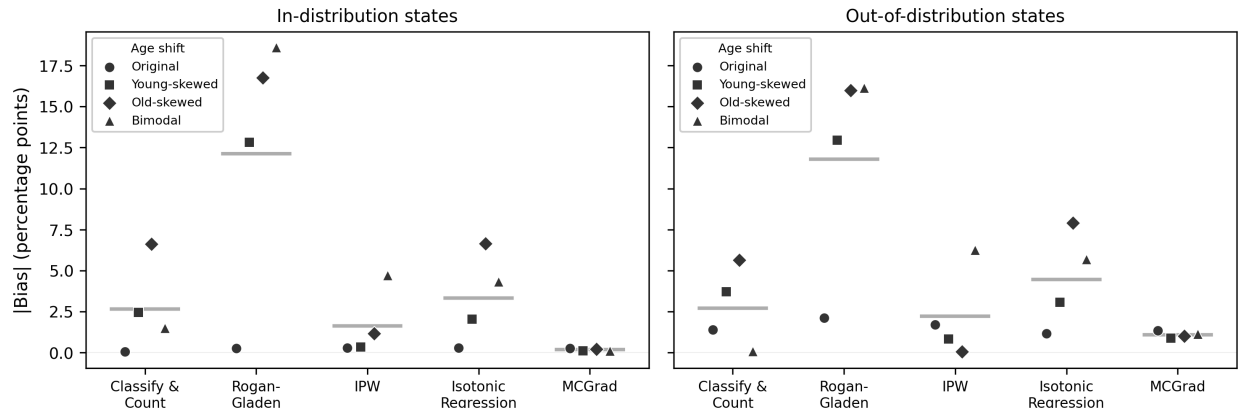


Figure 3. Absolute prevalence bias (percentage points) for the ACS employment benchmark, by method and age-shift scenario, for in-distribution (left) and out-of-distribution (right) states. Marker shape denotes the synthetic age distribution; horizontal lines are per-method means. MCGrad is near-unbiased across all in-distribution scenarios (including the bimodal shift that defeats IPW) and degrades only modestly out of distribution; Rogan-Gladen and isotonic regression grow with the age shift.

The pattern matches the simulation and CAP results (Figure 3; full numbers in SI Table S1). Rogan-Gladen fails by 12–19pp and SLD comparably; Classify & Count and isotonic regression show moderate but growing bias (up to 8pp); IPW is good on simple shifts (≤ 1.2 pp) but fails on the bimodal shift (+4.7pp in-distribution, +6.3pp out-of-distribution) where the density ratio is hard to model. Multicalibration achieves ≤ 0.27 pp bias across all in-distribution scenarios, including the bimodal shift, and degrades only modestly out of distribution (0.88–1.35pp), reflecting geographic shift along a dimension the calibration set did not span. Across both applications the story is consistent: multicalibration is near-unbiased when the target’s features lie within the calibration support and degrades predictably when they do not: the scope condition has visible, interpretable bite rather than silent failure.

Discussion

Estimating how prevalent a category is across populations is among the most common uses of text-as-data, and it is more fragile than current practice acknowledges. Discriminative validation certifies that a coder separates classes; it is silent about whether prevalence estimates transport. We have shown that the property that makes them transport is calibration conditional on features, which we obtain via multicalibration, and that this property turns a validated coder into a measurement instrument that can be calibrated once and applied to many populations. The result connects two literatures that developed apart: quantification, which corrected aggregate error rates but never engaged feature-conditional calibration, and multicalibration, which studied individual-level prediction and fairness but not population-level inference.

The central practical payoff is target-independence. Importance weighting, design-based supervised learning, and prediction-powered inference all redo work for each target: density ratios or gold samples drawn anew per population. A multicalibrated instrument is calibrated once and deployed without target-specific estimation, the property comparative and over-time research most needs. That advantage is bought with an assumption: the calibration features must capture the shift, and targets must lie in support. We view stating this scope condition, and showing its bite through

honest out-of-support results, as a strength rather than a caveat to bury.

Which tool to use is a choice about where labeled data comes from. If you can draw a gold-label probability sample *from each target* you analyze, design-based supervised learning (Egami et al. 2023) or prediction-powered inference (Angelopoulos et al. 2023) are efficient and require no covariate-shift assumption; prefer them. If instead you must measure across *many* targets without per-target hand-coding (the comparative-agendas situation, where re-coding thousands of documents per new country is infeasible), multicalibration provides a single reusable instrument, at the cost of the ignorability/support assumption. ReadMe-family quantifiers (Hopkins and King 2010; Jerzak et al. 2023) are the right choice under label shift (the prior changes, the class-conditional language does not), but fail under covariate shift (SI Section S7); multicalibration is the right choice under covariate shift.

Several limitations follow. First, the guarantee holds only along calibrated dimensions; novel feature values degrade estimates, so calibration data must be designed to span anticipated variation. Second, multicalibration requires labeled calibration data; in zero-shot settings these labels reintroduce the hand-coding that LLMs were meant to avoid, though far less of it and only once. When per-target gold labels are available, design-based methods are the better tool and the two approaches compose. Third, the covariate-shift scope condition (stated in the setup) can fail: under mixtures it holds only approximately, and under concept drift no purely statistical correction substitutes for new labels. Finally, our applications use settings with measurable ground truth; practitioners often lack it, which makes the support diagnostics above all the more important. The accuracy and AUC that text-as-data research reports remain necessary, but for valid prevalence estimates across contexts, they are not sufficient.

Data availability. Replication code is available at https://github.com/facebookresearch/multicalibrated_llm_measurement and will be deposited in the Political Analysis Dataverse on acceptance. ACS data is public via *folktables*; CAP data is public at <https://www.comparativeagendas.net>. Claude Opus inference outputs for all 30,000 main-text documents and Llama 3.3 70B outputs for all 105,000 documents used in SI Section S2 are included in the replication materials. A disclosure of generative-AI use in preparing this manuscript is provided in SI Section S8.

Competing interests. The authors declare no financial competing interests. In the interest of full disclosure, several authors are affiliated with Meta Platforms, and a subset are authors of the MCGrad algorithm used here; the open-source implementation is maintained by their team. The conceptual claim and the empirical guarantee are algorithm-agnostic, and any multi-accuracy or multicalibration fitter can be substituted for MCGrad. The open-weight Llama 3.3 70B replication (SI Section S2) shows the method does not depend on proprietary model access.

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